

guía (4)

obs

$$P(|x| \leq y) = P(-y \leq x \leq y) = F_X(y) - F_X(-y)$$

Función de V.A. "cambio de variable"

- sea X una V.A. / $X: \Omega \rightarrow \mathbb{R}$

$$\text{Im}(X) = R(X) = X(\Omega) \subset \mathbb{R}$$

- sea $g: D \subset \mathbb{R} \rightarrow \mathbb{R}$ tal que $R(X) \subset D$.

$$Y = g(X).$$

IMP: ver el soporte de la densidad g' re de de dato.

* método de eventos equivalentes

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

si queremos $f_Y(y) \rightarrow$ derivar

$$g^{-1}(y) = g(x)$$

teorema

$\hookrightarrow$ si X es abs continua, $Y = g(X)$ con g con derivada no nula $\rightarrow Y$ abs cont.

$$f_Y(y) = \frac{f_X(x)}{|g'(x)|} \Big|_{x=g^{-1}(y)}$$

g debe ser inyectiva (estrictamente creciente o decreciente).

Si es inyectiva acoluta

$\rightarrow$ parte g en funciones $\rightarrow f_Y(y) = \sum_{i=1}^k \frac{f_X(x_i)}{|g'(x_i)|} \Big|_{g^{-1}(y)=x_i}$

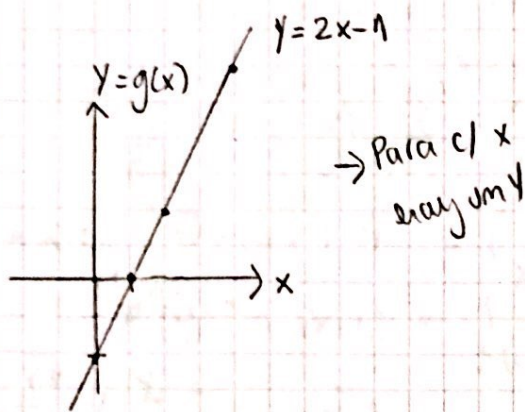
GUÍA 4

4.1) X , v.a discreta valores $\left\{ \frac{k}{8} : k=0, \dots, 8 \right\}$

$$P_X(x) = \frac{1}{36} x.$$

$$IAX = R(X) = \left\{ \frac{k}{8} : k=0, \dots, 8 \right\}$$

a) $Y = 2X - 1$



$$g(x) = 2x - 1$$

$$P_Y(2x-1=y) = \sum P_X\left(\frac{y+1}{2}=x\right)$$

$$IAY = g(IAX) = \left\{ 2 \cdot \frac{k}{8} - 1, k=0, \dots, 8 \right\}$$

$$\begin{aligned} 2x-1=y \\ x=\frac{k}{8} \end{aligned} = \left\{ -1, -3/4, \dots, 1 \right\}$$

$$P_Y(y) = \frac{1}{36} \left(\frac{y+1}{2} \right)$$

$$P_Y(y) =$$

$$1/36$$

$$0$$

$$1/288$$

$$1/144$$

$$1/96$$

$$1/72$$

$$5/288$$

$$3/144$$

$$7/288$$

$$y=1 \rightarrow k=8$$

$$y=-1 \rightarrow k=0$$

$$y=-3/4 \rightarrow k=1$$

$$y=-1/2 \rightarrow k=2$$

$$y=-1/4 \rightarrow k=3$$

$$y=0 \rightarrow k=4$$

$$y=1/4 \rightarrow k=5$$

$$y=1/2 \rightarrow k=6$$

$$y=3/4 \rightarrow k=7$$

X v.a **Discreta**

$R(X) \rightarrow$ rango x : cont de atomos

$Y = g(X)$ v.a discreta

$$R(Y) = \{ Y = g(X), X \in R(X) \}$$

$$P_Y(y) = P(Y=y) = P(g(X)=y) =$$

$$\sum P_X(x) =$$

$$x/y=g(x)$$

lo pongo en función de x , que es la que tengo el dato

si X tiene un solo valor no hace falta $\sum$

ej: hago $\sum$ cuando $y=1 \rightarrow x=x^2 \rightarrow x=1$
 $x=1$

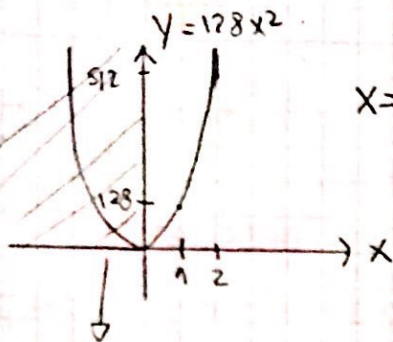
b) $Y = 128 X^2$

Para y hay 2 $x \rightarrow$ solo tomo del, sigue siendo impec.

$$X = \left\{ 0, \frac{1}{8}, \frac{2}{8}, \dots, 1 \right\}$$

$$Y = \{ 0, 2, 8, 18, 32, 50, 72, 98, 128 \}$$

$$X = \frac{k}{8}$$



como me
da los
valores de
 $X > 0$, solo
tomo lado
derecho

$$P(Y=Y) = P(Y=128X^2) = \sum P_X(X=\pm\sqrt{\frac{Y}{128}})$$

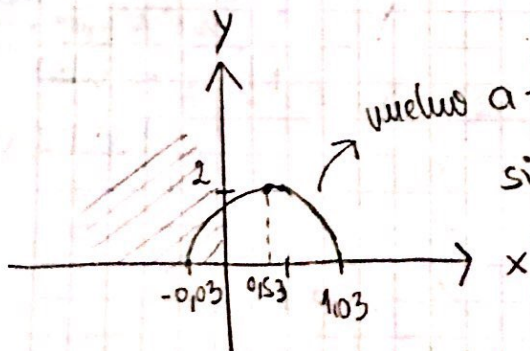
$$= \sum \frac{1}{36} \left(\pm\sqrt{\frac{Y}{128}} \right) = P\left(X = -\sqrt{\frac{Y}{128}} \cup X = +\sqrt{\frac{Y}{128}}\right)$$

descontando

$$= P\left(\frac{\sqrt{Y}}{\sqrt{128}}\right)$$

$P(Y=Y) =$	0	$Y=0$	
	$1/288$	$Y=2$	
	$1/144$	$Y=8$	
	$1/96$	$Y=18$	$1/48$ $Y=72$
	$1/72$	$Y=32$	$1/288$ $Y=98$
	$5/288$	$Y=50$	$1/36$ $Y=128$

c) $Y = -64X^2 + 64X + 2 \rightarrow X = \frac{k}{8} \rightarrow 0 < k < 1$



vuelvo a tener la parte > 0 , pero sea
si en algunos partes hay 2 valores
de X para cada Y .

max X

$X=0$ $Y=2$	$X=\frac{1}{8}$ $Y=9$	$X=\frac{2}{8}$ $Y=14$	$X=\frac{3}{8}$ $Y=17$	$X=\frac{4}{8}$ $Y=18$
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$$A_Y = \{ 2, 9, 14, 17, 18 \}$$

los 5 valores
no se suman.

$X=\frac{5}{8}$ $Y=17$	$X=\frac{6}{8}$ $Y=14$	$X=\frac{7}{8}$ $Y=9$	$X=\frac{8}{8}$ $Y=2$
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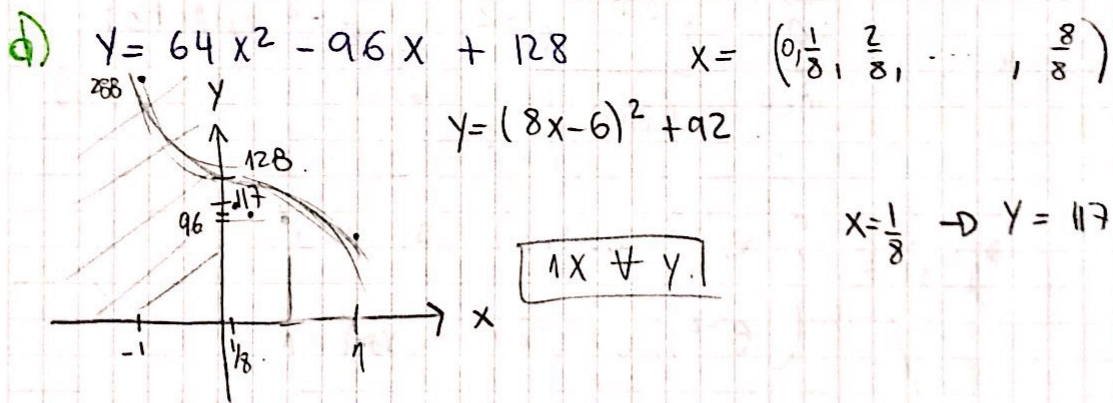
$$P_Y(y) = \frac{1}{36} (x)$$

$$x = \frac{-64 + \sqrt{64^2 + 4 \cdot 64(2-y)}}{2(-64)}$$

$$P_Y(y) = \begin{cases} P_X(0) + P_X(1) & y = 2 \\ P_X(1/8) + P_X(7/8) & y = 9 \\ P_X(2/8) + P_X(6/8) & y = 14 \\ P_X(3/8) + P_X(5/8) & y = 17 \\ P_X(4/8) & y = 18 \end{cases}$$

$$P_X(0) + P_X(1) = \frac{1}{36} \cdot 0 + \frac{1}{36} \cdot 1 = \frac{1}{36} \quad \text{... si go con los cuernos}$$

$\downarrow$ $x=0$ $\downarrow$ $x=1$
 $y=2$ $y=9$



$$P_Y(y) = P(x = [\sqrt{y-92} + 6] \cdot \frac{1}{8})$$

$$P_Y(y) = \frac{1}{36} (x)$$

$$P_Y(y) = \begin{cases} 1/128 & y = 117 \rightarrow x = 1/8 \\ 1/24 & y = 128 \rightarrow x = 0 \\ 1/36 & y = 96 \rightarrow x = 1 \end{cases}$$

(4.2)

 $X \sim P(2)$

discreta

$$\text{Poisson} = \frac{\mu^x e^{-\mu}}{x!}$$

$$Y = \left| \sin\left(\frac{1}{2}\pi x\right) \right| = \begin{cases} 0 & x=2m \\ 1 & x=2m+1 \end{cases}$$

↓
tenyo zopuulus
↳ par o impar

$$P_Y(Y=0) = P(X=2m) = \sum_{m=0}^{\infty} P(X=2m) = \sum_{m=0}^{\infty} \frac{2^{2m} e^{-2}}{(2m)!}$$

$$= e^{-2} \cosh(2)$$

$$P_Y(Y=1) = P(X=2m+1) = \sum_{m=0}^{\infty} P(X=2m+1) =$$

$$\sum_{m=0}^{\infty} \frac{2^{2m+1} e^{-2}}{(2m+1)!} = e^{-2} \sinh(2)$$

$$P_Y(y) = \begin{cases} e^{-2} \cosh(2) & \text{si } y=0 \\ e^{-2} \sinh(2) & \text{si } y=1 \end{cases}$$

si X es abs continua.

ej' si $y=x^2$.

$$F_Y(y) = P(Y \leq y) = P(g(x) \leq y) =$$

$$= P(x^2 \leq y) = P(x \leq \sqrt{y}) =$$

con $1 \leq y \leq 5$.

$$= F_X(\sqrt{y}) =$$

si queremos densidad derivamos

$$f_Y(y) = \frac{f_X(x)}{g'(x)}$$

con $x = g^{-1}(y)$

4.3

va. continua

$$f_x(x) = \frac{12x}{\pi^2 \cdot (e^x + 1)} \quad \text{II } \{x > 0\}$$

OBS:

$$y = ax + b$$

$$f_Y(y) = \frac{1}{a} f_x\left(\frac{y-b}{a}\right)$$

a) $y = ax + b \quad (a \neq 0, b \in \mathbb{R})$

si usara el de equivalentes

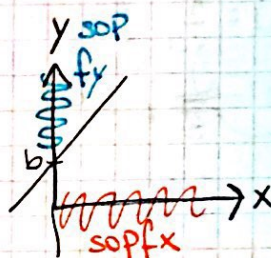
↓
muy complicado

$$f_Y(y) = \frac{f_x(x)}{g'(x)} = \frac{1}{a} f_x\left(\frac{y-b}{a}\right)$$

↓
 $f_X = \int_{\mathbb{R}^2} f_X dx \dots$

$$y = g(x) = ax + b$$

$$g'(x) = a$$



$$f_Y(y) = \frac{f_x(x)}{g'(x)} \Big|_{x=g^{-1}(y)}$$

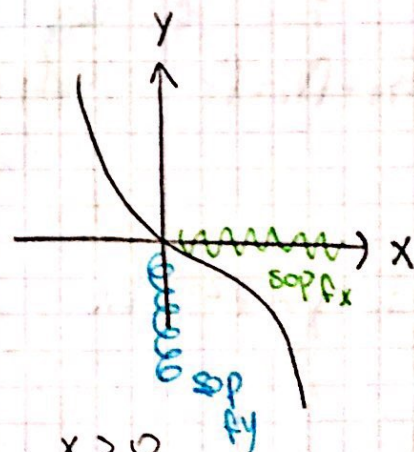
$$= f_x(x) \cdot \left(\frac{dg^{-1}(y)}{dy} \right)$$

$$f_Y(y) = \frac{1}{a} \left[\frac{12 \left(\frac{y-b}{a} \right)}{\pi^2 \left(e^{\left(\frac{y-b}{a} \right)} + 1 \right)} \right] \quad \text{II } \left\{ \frac{y-b}{a} > 0 \right\}$$

$\boxed{y > b}$

b) $y = -x^3 \rightarrow x = \sqrt[3]{-y}$

$$f_Y(y) = \frac{1}{-3(y^{1/2})} \left[\frac{12 (\sqrt[3]{-y})}{\pi^2 (e^{\left(\frac{-y}{\sqrt[3]{-y}} \right)} + 1)} \right] \quad \text{II } \{y < 0\}$$



$$g(x) = -x^3$$

$$g'(x) = -3x^2 = -3(-y)^{2/3}$$

OBS!

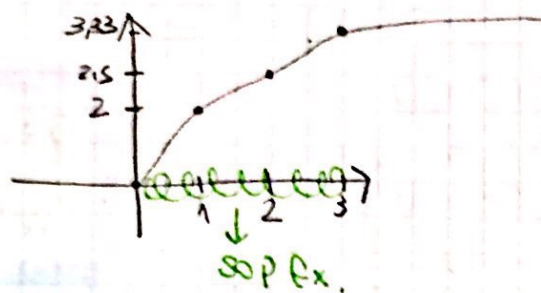
si $x \in \mathbb{R}^2$,
teniendo que pasar
a y en 2

Por $x < 0 \rightarrow \sqrt[3]{y}$
 $x > 0 \rightarrow \sqrt[3]{-y}$

$$\begin{aligned} x &> 0 \\ \sqrt[3]{-y} &> 0 \\ -y &> 0 \\ \boxed{y < 0} \end{aligned}$$

c)

$$y = x + x^{-1} = x + \frac{1}{x} = \frac{x^2 + 1}{x} =$$



$$xy = x^2 + 1$$

$$0 = x^2 - xy + 1$$

pongo a x en función de y

$$\begin{cases} x_1 = \frac{y + \sqrt{y^2 - 4}}{2} \\ x_2 = \frac{y - \sqrt{y^2 - 4}}{2} \end{cases}$$

$$y' = g'(x) = \frac{dy(x)}{dx}$$

$$x = g^{-1}(y)$$

$$x' = g^{-1}(y) = \frac{dx(y)}{dy}$$

$$g'(x) = \frac{1}{g^{-1}(y)}$$

Por la regla
de la derivada



$$f_{y1} + f_{y2} = f_y$$

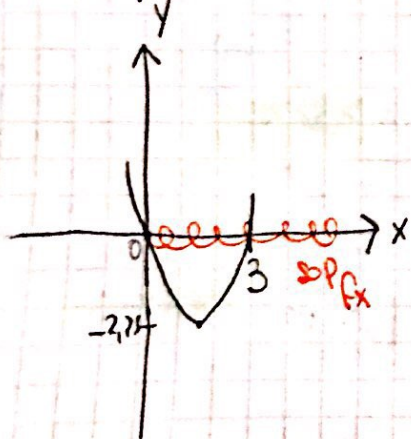
$$f_y(y) = f_x \cdot \frac{dx(y)}{dy}$$

como siempre...

d)

$$y = x^2 - 3x$$

$$0 = x^2 - 3x - y$$



$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{3 \pm \sqrt{9 - 4(-y)}}{2} = \frac{3 \pm \sqrt{9 + 4y}}{2} \begin{cases} x_1 \\ x_2 \end{cases}$$

$$g_1' = \dots$$

$$g_2' = \dots$$

$$f_y = f_{y1} + f_{y2}$$

$\downarrow$ $\downarrow$
 como f_{x1} f_{x2}

025

4.4

$$X \sim U(-\pi/2, \pi/2)$$

$$f_X(x)$$

El ángulo θ y
tangente x

x es mi y
de siempre
 θ es mi x de
siempre

$(X, 0)$ = punto intersección entre vertical $y=0$ y el eje x

$$\tan \theta = \frac{OP = x}{OQ = 1} \rightarrow \boxed{x = \tan \theta}$$

$$f_\theta = \frac{1}{\pi}$$

$$\begin{aligned} F_X(x) &= P(X \leq x) = P(\tan \theta \leq x) = P(\theta \leq \arctan x) \\ &= F_\theta(\arctan x) \end{aligned}$$

integrar
↓ f_θ

$$F_\theta(\theta) = \frac{\theta + \frac{\pi}{2}}{\frac{\pi}{2} + \frac{\pi}{2}} = \frac{\theta + \frac{\pi}{2}}{\pi} = \frac{\theta}{\pi} + \frac{1}{2}$$

$$f_\theta = \frac{1}{b-a}$$

$$\theta = \arctan(x)$$

$$\frac{d\theta}{dx} = \frac{1}{1+x^2}$$

$$F_\theta(\arctan(x)) = \frac{1}{2} + \frac{\arctan x}{\pi} = F_X(x)$$

derivar

$$\boxed{f_X(x) = F_X'(x) = \frac{1}{\pi} \left(\frac{1}{1+x^2} \right)}$$

4.5 La fase ϕ de un generador eléctrico, va $\sim U(-\pi, \pi)$

a) $c = \cos \phi$

$$(\arccos(x))' = \frac{-1}{\sqrt{1-x^2}}$$

$$\phi \sim U(-\pi, \pi)$$

integrar
↓

$$f_\phi(\psi) = \frac{1}{2\pi} \mathbb{1}_{\{-\pi < \psi < \pi\}}$$

$$F_\phi(\psi) = \frac{\psi + \pi}{2\pi} \mathbb{1}_{\{-\pi < \psi < \pi\}} + \mathbb{1}_{\{\psi \geq \pi\}}$$

derivar
↑

$$F_C(c) = P(\cos \theta \leq c) = P(|\theta| \geq \arccos c) =$$

$$P(\arccos(c) \leq \Phi \leq -\arccos(c)) =$$

$$\overset{\text{"4"}}{F\Phi(-\arccos(c))} - F\Phi(\arccos(c)) =$$

$$- \frac{\arccos(c) + \pi}{2\pi} - \frac{\arccos(c) + \pi}{2\pi} = - \frac{\arccos(c)}{\pi}$$

→ exemplo 30 em 10
F.i. 100% probab.

$$[F_c(c) = - \frac{\arccos(c)}{\pi} \quad || \quad \{-1 < c < 1\}]$$

↓
derivate

$$[f_c(c) = \frac{1}{\pi \sqrt{1-c^2}} \quad || \quad \{-1 < c < 1\}]$$

b) $P(|c| < 0.5) = P(-0.5 < c < 0.5) =$

$$F_c(0.5) - F_c(-0.5) = \text{evalua em } c = \dots$$

$$= - \frac{\pi}{3\pi} - \frac{-2\pi}{3\pi} = \boxed{\frac{1}{3}}$$

OBS

$$P(|c| < x) = P(-x < c < x) =$$

$$F_c(x) - F_c(-x) =$$

4.6 Lucas llega (7:05, 7:50)

$$L \sim U(7:05, 7:50)$$

Subte $\sim U(0, 15)$ a partir de los 6.

6:00, 6:15, 6:30, 6:45, 7:00, 7:15, 7:30, 7:45, 8:00

Lucas



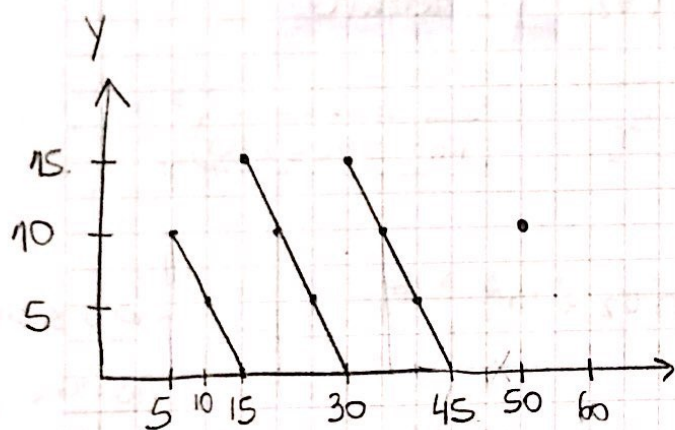
en el tiempo que Lucas esta llegando

3 subtes

X: hora de llegada de Lucas

$X \sim U(5, 50)$, mín.

Y: tiempo que espera



$$f_Y = \frac{f_X}{|g'(x)|} \Big|_{Y=g(x)} = \frac{1}{45} \mathbb{1}_{\{ \text{intervalo} \}}$$

$$Y = 15 - X$$

$$X = 15 - Y$$

$$g'(x) = -1$$

$$|g'(x)| = 1$$

$$f_X(x) = \frac{1}{45} \mathbb{1}_{\{ 5 < x < 50 \}}$$

$$Y = g(x) = \begin{cases} 15 - X & 7:05 \leq X \leq 7:15 \rightarrow 0 \leq Y \leq 10 \\ 30 - X & 7:15 < X \leq 7:30 \rightarrow 0 \leq Y \leq 15 \\ 45 - X & 7:30 < X \leq 7:45 \rightarrow 0 \leq Y \leq 15 \\ 60 - X & 7:45 < X \leq 7:50 \rightarrow 10 \leq Y \leq 15 \end{cases}$$

del $g(x) = f_X(x)$

$$\frac{1}{45} 0 \leq Y \leq 10 + \frac{1}{45} 10 \leq Y \leq 15. \text{ (por a pinto)}$$

$$f_Y(y) = \frac{1}{45} \mathbb{1}_{\{ 0 \leq Y \leq 10 \}} + \frac{1}{45} \mathbb{1}_{\{ 0 \leq Y \leq 15 \}} + \frac{1}{45} \mathbb{1}_{\{ 0 \leq Y \leq 15 \}} + \frac{1}{45} \mathbb{1}_{\{ 10 \leq Y \leq 15 \}}$$

$$[f_Y(y) = \frac{1}{15} \mathbb{1}_{\{ 0 \leq Y \leq 10 \}} + \frac{1}{15} \mathbb{1}_{\{ 10 \leq Y \leq 15 \}}]$$

→ (ARBITRA (P. 65)).

(4.7)

$$V_1 \sim U(180, 220)$$

Tabla

$$f(V_1) = \frac{1}{40} \mathbb{1}_{\{180 \leq V_1 \leq 220\}}$$

$$V_2 = g(V_1) = \frac{V_1 - 190}{20} \mathbb{1}_{\{190 \leq V_1 \leq 210\}} + \mathbb{1}_{\{210 < V_1\}}$$

$$V_2 = g(V_1)$$

$$g(V_1) = \begin{cases} \frac{V_1 - 190}{20} & 190 \leq V_1 \leq 210 \rightarrow g_1 \\ 1 & 210 < V_1 \rightarrow g_2 \end{cases}$$

integral $f(V_1)$ → $F(V_1) = \int_{180}^{V_1} f(V_1) dV_1 = \frac{1}{40} V \Big|_{180}^{V_1} =$

$$F(V_1) = \frac{V_1 - 180}{40} \mathbb{1}_{\{180 \leq V_1 \leq 220\}}$$

$$\begin{aligned} F_{V_2}(v_2) &= P(V_2 \leq v_2) = P(g(V_1) \leq v_2) = P(V_1 \leq g^{-1}(v_2)) \\ &= F_{V_1}(g^{-1}(v_2)) \rightarrow \text{TEOREMA} \end{aligned}$$

$$g(V_1) = V_2$$

$$g_1(V_1) = \frac{V_1 - 190}{20} = V_2 \quad \boxed{190 \leq V_1 \leq 210}$$

$$\begin{aligned} V_1 &= 20V_2 + 190 & 190 \leq 20V_2 + 190 \leq 210 \\ & & 0 \leq V_2 \leq \frac{210 - 190}{20} \end{aligned}$$

$$\boxed{0 \leq V_2 \leq 1}$$

$$210 \leq V_1 \leq 220 \quad g_2(V_1) = 1 = V_2 \quad 210 < V_1$$

$$\boxed{V_2 > 1} \quad \leftarrow \text{signo en analogía}$$

$$g^{-1}(V_2) = g_1(V_1)$$

$$V_1 = 20V_2 + 190$$

$$F_{V_2}(v_2) = F_{V_1}(g_1^{-1}(v_2)) + F_{V_1}(g_2^{-1}(v_2)) =$$

$$\frac{V_1 - 180}{40}$$

$$F_{V_2}(v_2) = \frac{20V_2 + 190 - 180}{40} \mathbb{1}_{\{V_2 \in (0, 1)\}} +$$

$$g_1(V_1) = V_2$$

$$\mathbb{1}_{\{V_2 > 1\}}$$

4.8

X: duración llamada va.
 $X \sim \exp(1/8)$

$$1/\lambda = E = 8 \text{ min.}$$

$$\lambda = 1/8.$$

$$f_X(x) = \frac{1}{8} e^{-1/8 x} \mathbb{1}(x > 0)$$

se factura 1 pulso c/ 2 min

cantidad de c facturados por llamada $\leftarrow P(C=c)$

C = "cont de pulsos facturados por llamada"

$$C = \begin{cases} 1 & 0 \leq X < 2 \\ 2 & 2 \leq X < 4 \\ 3 & 4 \leq X < 6 \\ \vdots & \vdots \end{cases}$$

evento elemental
 $\downarrow$

C

$$2c-2 \leq x \leq 2c$$

pongo a x en función de c

$$P(2c-2 \leq x \leq 2c) = F_X(2c) - F_X(2c-2) =$$

$$= \int_{2c-2}^{2c} f_X dx = \int_{2c-2}^{2c} \frac{1}{8} e^{-1/8 x} dx =$$

$$= \frac{1}{8} \frac{e^{-1/8 x}}{-1/8} \Big|_{2c-2}^{2c} = -e^{-x/8} \Big|_{2c-2}^{2c} =$$

$$= \left[-e^{-\frac{2c}{8}} + e^{-\frac{2c-2}{8}} \right]$$

CRUCIAL

$$\int e^{-1/8 x} dx = \frac{e^{-1/8 x}}{-1/8}$$

Recuerda (guía 2)

$$P(a \leq x \leq b) = F_X(b) - F_X(a) = \int_a^b f_X dx$$

Para **VECTORES** **aleatorios** (2 variables)

antes.
x y
g(x)=y

(x, y) v.a

$g: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$R(x, y) \subset D$

se define $z = g(x, y)$

$$F_z(z) = P(z \leq z) = P(g(x, y) \leq z) = P(g(x, y) \in (-\infty, z]) \\ = P((x, y) \in g^{-1}((-\infty, z]))$$

Ya por eventos complicados $\uparrow$, suelen ser el área.
muy complicados.

JACOBIANO

(x, y) v.a abs continua.

$g: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$\text{sop}(f_{xy}) \subset D$

$$g(x, y) = (g_1(x, y), g_2(x, y))$$

Defino

$$(u, v) = (\underbrace{g_1(x, y)}_u, \underbrace{g_2(x, y)}_v) = g(x, y)$$

Para hallar $f(u, v)$

$$f_{uv}(u, v) = f_{xy}(x(u, v), y(u, v)) \cdot \left| \frac{d(x, y)}{d(u, v)} \right|$$

o

$$f_{uv}(u, v) = \frac{f_{xy}(x(u, v), y(u, v))}{\left| \frac{d(u, v)}{d(x, y)} \right|}$$

$$\begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$\begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} \\ \frac{dv}{dx} & \frac{dv}{dy} \end{vmatrix}$$

donde $(x(u, v), y(u, v)) = g^{-1}(u, v)$

Nota = ver soporte de u, v .

si son inyectivas **localmente** (como antes)

$$f_{uv}(u,v) = \sum_{i=1}^m f_{xy}(x(u,v), y(u,v)) \left| \frac{d(x,y)}{d(u,v)} \right| \mathbb{1}_{\{u,v \in D\}}$$

↓
densidad por partes

$$g^{-1}(u,v) = (x(u,v), y(u,v))$$

SOP (x,y) & valores g' puede tomar el vector en $\mathbb{R}^2$.

si **$y=g(x)$**

g estrictamente **creciente** $\rightarrow F_y(y) = F_x(g^{-1}(y))$

g estrictamente **decreciente** $\rightarrow F_y(y) = 1 - F_x(g^{-1}(y))$

$U = \min(x,y)$

$$\begin{cases} x & \text{si } x \leq y \\ y & \text{si } y \leq x \end{cases}$$

$V = \max(x,y)$

$$\begin{cases} x & \text{si } x \geq y \\ y & \text{si } y \geq x \end{cases}$$

4.9) a) $U = x \quad V = x + y$

muestra tabla

u/v	-4	-3	-2	-1	0	1	2	3	4
-2	1/16	1/16	0	1/8	1/8	0	0	0	0
-1	0	1/16	1/16	0	1/8	1/16	0	0	0
1	0	0	0	0	0	0	1/16	1/16	0
2	0	0	0	0	0	0	0	1/16	1/16

$$f_{UV} = \frac{f_{xy}}{J} = \frac{f_{xy}}{1} = f_{xy}$$

$$\left| \frac{d(u,v)}{d(x,y)} \right| = \begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} \\ \frac{dv}{dx} & \frac{dv}{dy} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1$$

$$(2, -2) = (2, 0)$$

$$(2, -1) = (2, 1)$$

$$(2, 1) = (2, 3)$$

$$(2, 2) = (2, 4)$$

$xy \quad uv$

b) $U = \min(x, y)$

$V = \max(x, y)$

X	Y	U	V
(-2, -2)		(-2, -2)	
(-2, -1)		(-2, -1)	
(-2, 1)		(-2, 1)	
(-2, 2)		(-2, 2)	

U/V	-2	-1	1	2
-2	1/16	1/16	1/8	1/8
-1	0	1/16	1/8	1/8
1	0	0		
2	0	0		

$$\begin{pmatrix} -1 & -2 \\ -1 & -1 \\ -1 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} -2 & -1 \\ -1 & -1 \\ -1 & 1 \\ -1 & 2 \end{pmatrix}$$

1-2 $\rightarrow$ -2-1
sign...
...

c) $U = x^2 + y^2$ $V = \frac{y}{x}$
mismo gl'antes

$$J = \begin{vmatrix} 2x & 2y \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}$$

4.11 $z_1, z_2 \sim (0, 1)$ z_1, z_2 ind. $\mu=0$ $\sigma=1$

a) f_{UV}, f_U, f_V

1) $U = z_1 + z_2$ $V = z_1 - z_2$

$f_{z_1} = \frac{1}{\sqrt{2\pi}} e^{-\frac{z_1^2}{2}}$

$f_{z_2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{z_2^2}{2}}$

preserv indep.

$f_{z_1, z_2} = f_{z_1} f_{z_2}$

$f_{UV} = f_{z_1, z_2} \left(z_1 = \frac{U+V}{2}, z_2 = \frac{-U+V}{2} \right)$

$$J = \begin{vmatrix} \frac{\partial U}{\partial z_1} & \frac{\partial U}{\partial z_2} \\ \frac{\partial V}{\partial z_1} & \frac{\partial V}{\partial z_2} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -1 - 1 = -2$$

$|J| = 2$

$z_2 = \frac{-U+V}{2}$

$$f_{z_1 z_2} = \frac{1}{2\pi} e^{-\frac{1}{2}(z_1^2 + z_2^2)}$$

$$\| (z_1, z_2) \| \in \mathbb{R}^2$$

$$f_{uv} = f_u \cdot f_v$$

$$\begin{aligned} & \left(-\frac{v}{2}\right)^2 + 2\left(-\frac{v}{2}\right)\left(\frac{v}{2}\right) + \left(\frac{v}{2}\right)^2 \\ & \left(\frac{v}{2}\right)^2 - 2\left(\frac{v}{2}\right)\left(-\frac{v}{2}\right) + \left(-\frac{v}{2}\right)^2 \end{aligned}$$

$$f_{uv} = \frac{1}{4\pi} e^{-\frac{1}{2} \left(\left(\frac{v+v}{2}\right)^2 + \left(\frac{v}{2} - \frac{v}{2}\right)^2 \right)}$$

$$f_u = \frac{1}{2\pi} e^{-\frac{1}{2} \left(\frac{v}{2}\right)^2} \quad f_v = \frac{1}{2\pi} e^{-\frac{1}{2} (v)^2}$$

chequear
cuentos
igual

$$U \sim N(0, 2)$$

$$V \sim N(0, 1)$$

$$2) \quad U = \cos \theta z_1 - \sin \theta z_2$$

$$z_1 \sim N(0, \cos^2 \theta) \quad V = \sin \theta z_1 + \cos \theta z_2$$

$$z_2 \sim N(0, \sin^2 \theta)$$

misma idea

$$U \sim N(0, 1)$$

$$V \sim N(0, 1)$$

$$3) \quad U = z_1^2 + z_2^2 \quad V = \frac{z_2}{z_1} \quad \text{tamb.}$$

$$U \sim \text{exp}(1/2) \quad V \sim \text{Cauchy}(0, 1)$$

b) son ind, pque si uno es min, depende del otro.

$$c) \quad P(z_1^2 + z_2^2 > 4) = P(U > 4) = 1 - P(U \leq 4) = 1 - e^{-4/2} \approx 0.86$$

$$F_U(4)$$

$$P(z_2 > \sqrt{3} z_1) = P\left(\frac{z_1}{z_2} > \sqrt{3}\right) = P(V > \sqrt{3}) =$$

$$1 - P(V \leq \sqrt{3}) = 1 - F_V(\sqrt{3}) =$$

IMP:

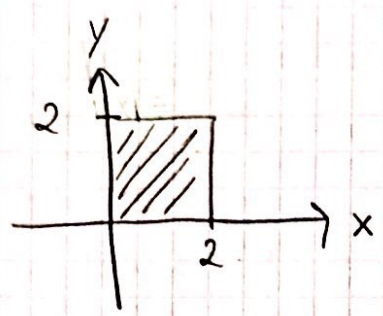
aunque al hacer cambio de variables,
quedan muchos distribuciones, queda
cambio (analizarlo)

NOTA

4.12
 mejor
 4.10 compita

$X_1, X_2 \sim (0,2)$ $\nearrow$ # estandar $\mu=0 \quad \sigma=2$
 independientes
 $U = \min(X_1, X_2) \quad V = \max(X_1, X_2)$

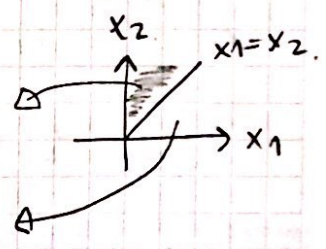
a) f_{UV}



$f_{X_1 X_2} = \frac{1}{\text{area}} \quad \parallel \{ X_1, X_2 \in (0,2)^2 \}$

area = 4

$$(U, V) = \begin{cases} (X_1, X_2) & \text{si } X_1 < X_2 \\ (X_2, X_1) & \text{si } X_1 > X_2 \end{cases}$$



$$\begin{aligned} F_{UV}(\mu, \nu) &= P((U, V) \leq (\mu, \nu)) = P(\min(X_1, X_2), \max(X_1, X_2) \leq \mu, \nu) \\ &= P(U \leq \mu, V \leq \nu, X_1 < X_2) + P(U \leq \mu, V \leq \nu, X_2 < X_1) \end{aligned}$$

$$F_{UV}(\mu, \nu) = \begin{cases} 0 & \mu < 0, \nu < 0 \\ \frac{1}{2} \nu \mu - \frac{1}{4} \mu^2 & (\mu, \nu) \in (0,2) \\ 1 & \mu > 2, \nu > 2 \end{cases}$$

$\rightarrow$ no extend. mucho

mejor $\nearrow$

$f_{UV} = \int_1^2 \int_1^2 f_{XY} |J| =$

$|J| = 1$ en ambos casos

$$f_{UV} = \frac{1}{\sqrt{2\pi} \cdot 2} e^{-\frac{(\mu)^2}{2 \cdot 4}} + \frac{1}{\sqrt{2\pi} \cdot 2} e^{-\frac{\nu^2}{8}}$$

caso 1
 $\mu = x \quad \nu = y$

caso 2
 $\mu = y \quad \nu = x$
 $\parallel \{ (\mu, \nu) \in (0,2)^2 \}$

b) $W = V - U = \max - \min$

Solo para
exponenciales TEOREMA.

4.14

- $X \sim \exp(\lambda_1)$ • $Y \sim \exp(\lambda_2)$ • X, Y ind
- $U = \min(X, Y)$ • $W = V - U$
- $V = \max(X, Y)$ • $J = 1 \{U = X\} + 2 \{U = Y\}$

a) densidad de U .

$$F_U(u) = 1 - e^{-(\lambda_1 + \lambda_2)u}$$

derivo $\rightarrow$

$$f_U(u) = (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2)u} \quad \mathbb{1}\{u > 0\}$$

$$U \sim \exp(\lambda_1 + \lambda_2)$$

EL MÍNIMO ENTRE EXPONENCIALES INDEPENDIENTES,
sigue siendo una exponencial de parámetro $\lambda_1 + \lambda_2$

b) Función prob de J .

$$P(J=1) = \frac{\lambda_1}{\lambda_1 + \lambda_2} \rightarrow \text{el mínimo es } X$$

$$P(J=2) = 1 - \frac{\lambda_1}{\lambda_1 + \lambda_2} \rightarrow \text{el mínimo es } Y$$

c) densidad de $W = V - U$.

$$FW(0) = 0$$

$$FW(\infty) = 1$$

$$FW(w) = 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-\lambda_1 w} - \frac{\lambda_1}{\lambda_1 + \lambda_2} e^{-\lambda_2 w}$$

$$f_W(w) = \underbrace{\frac{\lambda_2 \lambda_1}{\lambda_1 + \lambda_2}}_p e^{-\lambda_1 w} + \underbrace{\frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}}_{1-p} e^{-\lambda_2 w}$$

El promedio entre
densidad X y Y = densidad
 $\max - \min$

$f_X(w)$
densidad X

$f_Y(w)$
densidad Y

d) MOSTRAR U e J SÖU INDEPENDENTES

U CONTINUA J DISCRETA

$$P(U > u, J=1) = \underbrace{\frac{\lambda_1}{\lambda_1 + \lambda_2}}_{P(J=1)} \underbrace{e^{-(\lambda_1 + \lambda_2)u}}_{P(U > u)} = P(J=1) P(U > u)$$

U e J SÖU INDEPENDENTES

e) U e W SÖU INDEPENDENTES
ambas CONTINUAS

$$f_{UW}(u, w) = f_U(u) f_W(w)$$

$$f_U(u) = (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2)u}$$

$$f_W(w) = \frac{\lambda_2 \lambda_1}{\lambda_1 + \lambda_2} e^{-\lambda_1 w} + \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2} e^{-\lambda_2 w}$$

$$f_{UV}(u, v) = \frac{f_{X,Y}}{|J_1|} + \frac{f_{X,Y}}{|J_2|}$$

$\downarrow$
si

$$f_{X,Y} = \frac{1}{2} \{0 < x < 1, 0 < y < 1\} +$$

2 intervalos

$$2 \left\{ \frac{1}{2} < x < 1, \frac{1}{2} < y < 1 \right\}$$

Página close

J y P independientes.

4.15 $X = J \sim \exp\left(\frac{5}{60}\right)$

$Y = P \sim \exp\left(\frac{10}{60}\right)$

J: tiempo entre llamadas de Juan.

$\lambda_1 = \frac{5}{60}$ $\lambda_2 = \frac{10}{60}$ λ_1, λ_2 en minutos

- antes de los 5 min no se atienden llamadas
- desde la última llamada, dejan pasar 5 minutos

Arrancan 10:00.

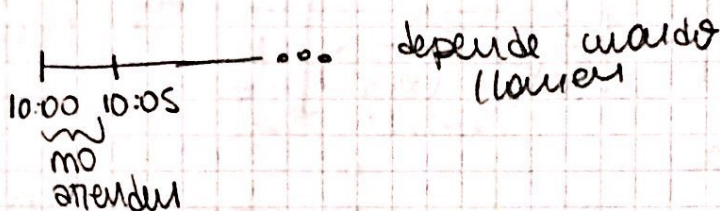
$$J = \begin{cases} 1 & \text{si } 1^{\text{er}} \text{ llamada Juan} \\ 2 & \text{si } 1^{\text{er}} \text{ llamada Pedro} \end{cases}$$

$U = X$
 $U = Y$

OBS.

$\lambda = \frac{5}{60}$
 $U \geq 5$
 o
 $\lambda = \frac{5}{60}$
 $U < \frac{5}{60}$

↓
X HORA



$U = 1^{\text{er}} \text{ llamada y mínimo}$

a) $P(U < 5) = 1 - P(U \geq 5) = 1 - F_U(5) =$

PROB 4.14

$U \sim \exp(\lambda_1 + \lambda_2)$

$= 1 - e^{-(\lambda_1 + \lambda_2)u}$

$= 1 - e^{-(\frac{15}{60}) \cdot 5}$

$P(U < 5) \approx 0.71$

1er llamada desde Juan

b) $P(J=1) = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{1}{3}$

c) $P(U < 5 | J=1) = \frac{P(J=1) \cap P(U < 5)}{P(J=1)} =$

↓
son indep.

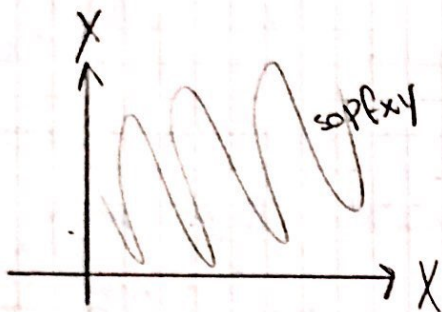
$= \frac{P(J=1) P(U < 5)}{P(J=1)} \approx 0.71$

4.16

X, Y ind $y \exp \lambda > 0$ ver en tabla que su soporte es de $[0, +\infty]$.

$$U = X + Y \quad V = \frac{X}{X + Y}$$

a)



$$\frac{du}{dx} = 1 \quad \frac{du}{dy} = 1$$

$$\frac{dv}{dx} = \frac{1 \cdot (x+y) - x \cdot (1)}{(x+y)^2}$$

$$\left| \frac{d(u,v)}{d(x,y)} \right| = \begin{vmatrix} 1 & 1 \\ \frac{y}{(x+y)^2} & \frac{-x}{(x+y)^2} \end{vmatrix}$$

$$\frac{dv}{dy} = \frac{-x \cdot (1)}{(x+y)^2}$$

por ser ind.
 $\downarrow$
 $f_{XY} = f_X f_Y$

$$\left| -\frac{x}{(x+y)^2} - \frac{y}{(x+y)^2} \right| = \left| \frac{-(x+y)}{(x+y)^2} \right| = \frac{x+y}{(x+y)^2}$$

$$f_{UV} = \frac{f_{XY}(x,y)}{\left| \frac{d(u,v)}{d(x,y)} \right|}$$

$\det J = \boxed{\frac{1}{x+y}} \rightarrow$ como se anula $\downarrow$ es biyectiva $\downarrow \exists g^{-1}$

$$X = U - Y$$

$$V = \frac{U-Y}{U-Y+Y} \Rightarrow 1 \cdot Y$$

$$X = U - U(1-V)$$

$$X = U - U + UV$$

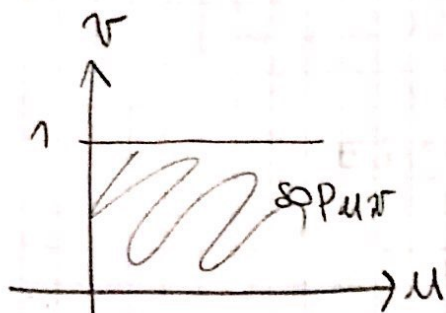
$$\boxed{X = UV}$$

$$V \cdot U - U = -Y$$

$$U - UV = Y$$

$$\boxed{U(1-V) = Y}$$

$$f_{UV} = \frac{\lambda e^{-\lambda x} \cdot \lambda e^{-\lambda y}}{\frac{1}{x+y}} = \frac{\lambda^2 e^{-\lambda(x+y)}}{1/x+y} \quad \text{on } \{x > 0, y > 0\}$$



$$x > 0 \rightarrow uv > 0$$

$$v > 0 \text{ o } u > 0$$

$$y = u(1-v) > 0$$

$$\begin{matrix} 1-v > 0 \\ 1 > v \end{matrix}$$

$$0 < v < 1$$

$$u > 0$$

↓
nuevos
parámetros.

$$f_{uv}(u, v) = \frac{\lambda^2 \cdot e^{-\lambda(uv + u(1-v))}}{\underbrace{(uv + u(1-v))}_u} =$$

$$f_{uv}(u, v) = u \cdot \lambda^2 e^{-\lambda u} \quad \begin{matrix} 0 < v < 1 \\ u > 0 \end{matrix}$$

como el soporte es
un rectángulo
no descarto que no
sean ind.

Para las marginales
integro sobre
el soporte

↓
 $f_u \rightarrow$ sobre v
como una u .

$f_v \rightarrow$ sobre u .

↓
ya que de
 f_u se ve todo
el soporte

$$f_u(u) = \int_0^1 \lambda^2 u e^{-\lambda u} dv$$

$$f_u(u) = \lambda^2 u e^{-\lambda u} \quad [u > 0]$$

$$f_v(v) = \int_0^{\infty} \lambda^2 u e^{-\lambda u} du$$

↓ por el soporte puedo
decir que se parece

se parece
a una
gamma →
↓
soporte

$$v = 2$$

$$U \sim \Gamma(2, \lambda)$$

↓
uniforme
[a, b]

$$V \sim U(0, 1)$$

b) $f_{uv} = f_u \cdot f_v$ ✓ son
ind

NOTA

4.17

(X, Y) punto aleatorio $\sim N(0, 1)$, x e y ind.

$$f_{xy} = \frac{1}{2\pi} e^{-1/2 (x^2 + y^2)} \quad \text{si } \{(x, y) \in \mathbb{R}^2\}$$

↓

f_x, f_y

$$X = \rho \cos \theta \quad Y = \rho \sin \theta$$

$$\text{Jac} \begin{vmatrix} \frac{dx}{d\rho} & \frac{dx}{d\theta} \\ \frac{dy}{d\rho} & \frac{dy}{d\theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -\rho \sin \theta \\ \sin \theta & \rho \cos \theta \end{vmatrix} = \rho$$

$$f_{\rho\theta}(\rho, \theta) = f_{xy}(x, y) \cdot |\text{Jac}|$$

$$= \frac{1}{2\pi} e^{-1/2 (\rho^2)} \cdot \rho \quad \text{si } \{\rho > 0, \theta \in (0, 2\pi)\}$$

$$\boxed{f_{\rho\theta}(\rho, \theta) = \frac{\rho}{2\pi} e^{-1/2 (\rho^2)} \quad \text{si } \{\rho > 0, \theta \in [0, 2\pi]\}}$$

$$f_{\rho}(\rho) = \int_0^{2\pi} f_{xy} d\theta = e^{-1/2 (\rho^2)} \quad \text{si } \{\rho > 0\}$$

$$f_{\theta}(\theta) = \int_0^{+\infty} f_{xy} d\rho = \int_0^{+\infty} \frac{\rho}{2\pi} e^{-1/2 (\rho^2)} d\rho$$

Como el
separ de θ es $(0, 2\pi) \rightarrow$ uniforme.

$$\downarrow$$
$$f_{\theta}(\theta) = \frac{1}{2\pi - 0} = \frac{1}{2\pi}$$

Comprobo q' sean ind.

4.19) $X, Y \text{ va } \sim \text{Unif inde.}$

$$X \sim U(1, 36)$$

$$Y \sim U(1, 4)$$

discreta = neue atomo

$$P_x(x) = \frac{1}{m} \rightarrow \text{msieur 36.}$$

$$\hookrightarrow U(m)$$

$$Z = g(X, Y) = X + Y$$

$$P(W=w) = P(X+Y=w)$$

[1, 36]
[1, m]
Sopont

$$P(X=1) = P(X=2) = \dots = P(X=36) = \frac{1}{36}$$

$$P(Y=1) = \dots = P(Y=4) = \frac{1}{4}$$

$$P(W=w) = \begin{cases} P(W=2) = P(X=1, Y=1) = \frac{1}{4} \cdot \frac{1}{36} = \frac{1}{144} \\ P(W=3) = P(X=1, Y=2) + P(X=2, Y=1) = \frac{1}{144} + \frac{1}{144} = \frac{2}{144} \\ \vdots \\ \text{y an con todos los variables} \\ P(W=m) = \frac{1}{36} \end{cases}$$

4.20) $L \sim \text{Poisson}(2)$
 $M \sim \text{Poisson}(8)$

$L, M \text{ ind.}$

DISCRETAS

IND

$$\begin{array}{l|l} M+L=m & L=1 \\ L=0 & M=m-1 \\ \hline M=m & \end{array}$$

$$\begin{aligned} a) P_{M+L}(m) &= P(M+L=m) = \sum_{l=0}^m P(L=l, M=m-l) = \\ &= \sum_{l=0}^m P(L=l) \cdot P(M=m-l) = \\ &= \sum_{l=0}^m \frac{e^{-2} 2^l}{l!} \cdot \frac{e^{-8} 8^{(m-l)}}{(m-l)!} \Rightarrow \end{aligned}$$

$$e^{-(10)} \sum_{l=0}^m \frac{2^l \cdot 8^{m-l}}{l! (m-l)!} \frac{m!}{m!}$$

$$\begin{aligned} M+L &= m \\ L &= l \\ M &= m-l \end{aligned}$$

$$M+L \sim \text{Po}(\underbrace{\mu_M + \mu_L}_{10})$$

$$\left. \begin{aligned} M &\sim \text{Po}(\mu_1) \\ L &\sim \text{Po}(\mu_2) \end{aligned} \right\} M+L \sim \text{Po}(\mu_1 + \mu_2)$$

Binomio de Newton

$$(a+b)^m = \sum_{i=0}^m \binom{m}{i} a^i b^{m-i}$$

$$b) P(M | (L+M)=10) = P(M=m | L+M=10)$$

$$= \frac{\sum_{m=0}^{10} P(M=m | L=10-m)}{P(L+M=10)}$$

$$\left. \begin{aligned} M &= m = m-l \\ L &= m-m \end{aligned} \right\}$$

$$= \frac{\frac{e^{-8} 8^m}{m!} \cdot \frac{e^{-2} 2^{(10-m)}}{(10-m)!}}{\frac{e^{-10} 10^{10}}{10!}} = \frac{10! 8^m 2^{10-m}}{m! (10-m)! 10^{10}}$$

$$= \frac{10!}{m! (10-m)!} \left(\frac{8}{10}\right)^m \left(\frac{2}{10}\right)^{10-m}$$

Grav: $M |_{L+M=10} \sim B\left(\begin{matrix} m=10 \\ p=8/10 \end{matrix}\right)$

$$M |_{L+M=m} \sim B\left(m, \frac{\mu_M}{\mu_M + \mu_L}\right)$$

↑ 10072

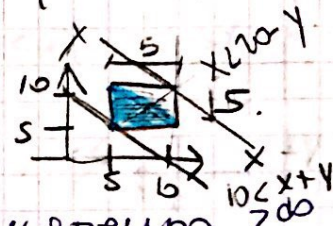
$$c) P(M > 2 | L+M=10) = \sum_{i=3}^{10} P_M | L+M=10(i^0)$$

$$1 - \sum_{i=0}^2 P_M | L+M=10 = 1 - \left[\binom{10}{0} \left(\frac{8}{10}\right)^0 \left(\frac{2}{10}\right)^{10} + \right.$$

$$\left. \binom{10}{1} \left(\frac{8}{10}\right)^1 \left(\frac{2}{10}\right)^9 + \binom{10}{2} \left(\frac{8}{10}\right)^2 \left(\frac{2}{10}\right)^8 \right] =$$

(4.21)

$$X, Y \text{ ind} \sim U(5, 10)$$



$$Z = Y + X.$$

X: tiempo 1º proceso y: tiempo 2º proceso

$$a) Z = X + Y$$

$$F_Z(z) = P(Z \leq z) = P(X+Y \leq z)$$

$$f_{XY} = \frac{1}{5 \times 5} = \frac{1}{25}$$

X+Y
↓
MIN=10
MAX=20

$$\Delta = \frac{bh/2}{25} = \frac{(z-10)^2}{50}$$

$$\nabla = \frac{(20-z)^2}{50}$$

$$F_Z(z) = \begin{cases} 0 & \text{si } z < 10 \\ \frac{(z-10)^2}{50} & \text{si } 10 \leq z \leq 15 \\ \frac{(20-z)^2}{50} & \text{si } 15 \leq z \leq 20 \\ 1 & \text{si } z \geq 20 \end{cases}$$

$$Z = X + Y$$

$$10 < Z < 20$$

$$10 < X + Y$$

$$10 - Y < X$$

$$X < 20 - Y$$

$$f_Z(z) = \frac{z-10}{25} \mathbb{1}_{\{10 \leq z < 15\}} + \frac{4}{5} - \frac{z}{25} \mathbb{1}_{\{15 \leq z < 20\}}$$

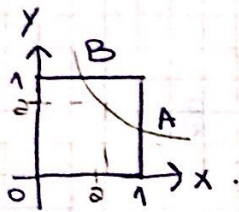
$$+ \mathbb{1}_{\{z < 10\}}$$

NOTA

b) el objeto se produce en menos de 16m/m
 $x+y < 16$

$$P(x+y \leq 16) = F_z(16) = -\frac{z^2}{50} + \frac{4xz}{5} - 7 = \frac{17}{25} = 0,68$$

$\uparrow$ $z=16$
 $\downarrow$ en mg
 en $15 < z < 20$



$A, B \sim U(0,1)$ ind.

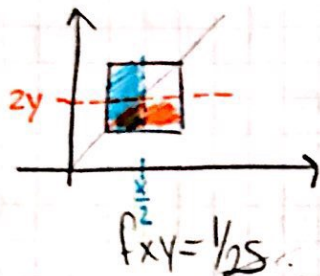
a) $f_R \Rightarrow P = A \times B$

$$F_R(r) = P(R \leq r) = P(A \times B \leq r)$$

$$A \square = A \square - A \square \quad \text{mínima.} \quad \hookrightarrow A = x \cdot L$$

4.24 $x, y, \text{ i.i.d. } \sim U(0, 5)$

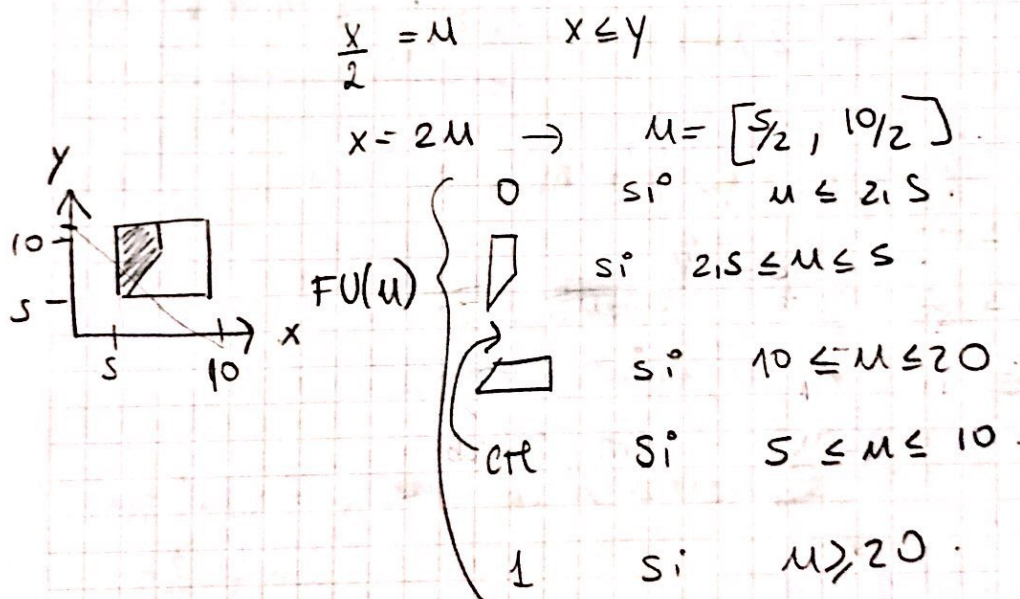
a) $U = \frac{x}{2} \mathbb{1}\{x \leq y\} + 2y \mathbb{1}\{x > y\} \rightarrow \text{densidad de } U.$



$$F_U(u) = P(U \leq u) = P(U \leq u, x \leq y) + P(U \leq u, x > y) =$$

$$P\left(\frac{x}{2} \leq u, x \leq y\right) + P(2y \leq u, x > y)$$

ver mi mano
soparte



denso halla $f_U(u)$

b) $V = \mathbb{1}\{x + y \leq 10\}$ $V = 0$ si $x + y > 10$
 $V = 1$ si $x + y \leq 10$

$$P(V=v) = \begin{cases} P(V=0) = P(x+y > 10) = 1/2 \\ P(V=1) = P(x+y \leq 10) = 1/2 \end{cases}$$

NOTA

4.25

$$f_x(x) = 2x \text{ on } \{0 \leq x \leq 1\}$$

$$f_y(y) = (2-2y) \text{ on } \{0 \leq y \leq 1\}$$

$$F(u) = F(x+y)$$

$$F_z(z) = P(x+y \leq z) = \int_{-\infty}^{+\infty} \int_{-\infty}^{z-x} f_{xy}(x,y) dy dx$$

$$f_z = \int_{-\infty}^{+\infty} f_{xy}(x, z-x) dx = \int_{-\infty}^{+\infty} f_x(x) f_y(z-x) dx$$

$$\begin{aligned} &= \int_{-\infty}^{+\infty} 2x (2-2(z-x)) dx = \int_0^1 4x(1-z) 2x^2 dx \\ &= 2x^2(1-z) - 2x^3 \Big|_0^1 = 1-2z \end{aligned}$$

$$[f_z(z) = 1-2z \text{ on } \{z = x+y\}]$$

4.24

100%

$f_{x|y}(x|y) = P$

4.25

4.24

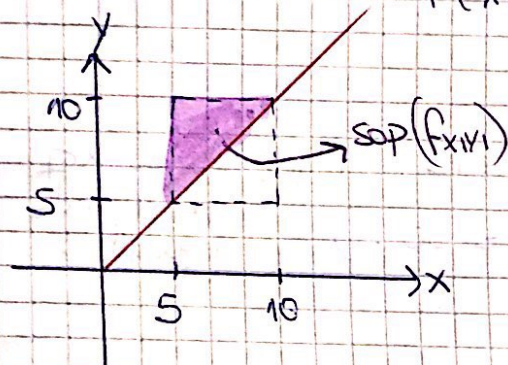
c) Hallar la función densidad de $X_1 + Y_1$ para $(x_1, y_1) = (x, y) \mid x < y$

TEORIA.

$f_{X|X \in B} = \frac{f_X}{P(X \in B)}$ llamo $z = x_1 + y_1$

↳ quiero la función densidad de z

$$P(X \in B) = \int_B f_X(x) dx \quad f_{X_1 Y_1}(x, y) = \frac{f_{X,Y}(x, y)}{P(X < Y)} \quad \begin{matrix} x < y < 0 \\ x < y \end{matrix}$$



$$P(X < Y) = \iint f_{X,Y}(x, y) dx dy$$

$$P(X < Y) = \iint \frac{1}{25} dx dy = \frac{1}{25} \text{ area}$$

al ser uniforme.

$X \sim U(5, 10)$
 $Y \sim U(5, 10)$ } indep.

base 5. altura 5.

$$\frac{5 \times 5}{2} = \frac{25}{2}$$

$$P(X < Y) = \frac{1}{25} \cdot \frac{25}{2} = \frac{1}{2}$$

$$f_{X,Y} = f_X f_Y =$$

$$= \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{25}$$

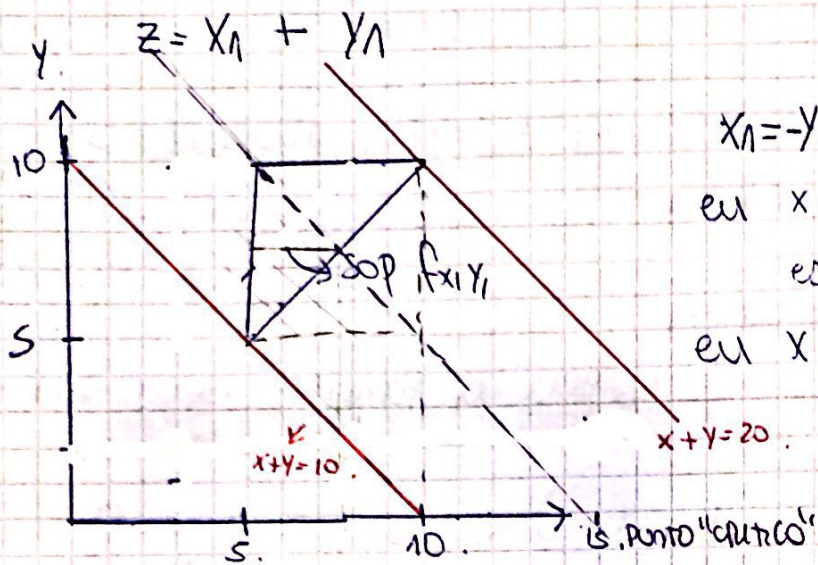
$$f_{X,Y} = \frac{1}{\text{area } 5} = \frac{1}{25} \rightarrow \text{same}$$

$$f_{X_1 Y_1}(x, y) = \frac{2}{25} \mathbb{1}_{\{5 < x < 10, 5 < y < 10, x < y\}}$$

PERO yo estoy hablando de z

$$z = x_1 + y_1$$

NOTA



voy a ir barriendo las zonas.

$$F_Z(z) = \begin{cases} 0 & z \leq 10 \\ ① & 10 < z \leq 15 \\ ② & 15 < z \leq 20 \\ 1 & z > 20 \end{cases}$$

los divido xq no cambia nada igual.

① $10 < z \leq 15 \quad P(M \leq z)$

$$\frac{(m-10)^2}{25}$$

$$F_{M|N \in B}(m) = \int \frac{f_{x,y}}{P(x \in B)} dt$$

② $1 - \frac{(20-m)^2}{25}$

$$= \frac{2}{25} \cdot \frac{1}{2}$$